

Partha Ghosh

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ppaatha

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I build LLM agents that sit on real building systems—not demo chatbots. At **Conneqted Agents** I work on **Optiify**: **LangGraph** tool-calling for bookings and HVAC faults, **MCP** tools for live point data, **Neo4j** equipment graphs, and **RAG** over site documents. Previously I worked on **RAG** chat, Bengali **STT/TTS**, and production **ML** at **REVE Systems**, including **GraphRAG**, **LightRAG**, multimodal and agentic **RAG** with **LangChain**.

EDUCATION

Shahjalal University of Science and Technology

Sylhet, Bangladesh

B.Sc. in Computer Science and Engineering

CGPA: 3.35 / 4.00

Feb 2018 – Mar 2023

WORK EXPERIENCE

ConneqtedAgents

Dec 2025 – Present • Remote

AI Engineer

- Built an in-process **LangGraph** agent for **Optiify** chat so building users can book after-hours access, inspect HVAC faults, and ask equipment questions in one conversation.
- Designed tool-calling workflows (booking, **FDD**, work orders, building context) with **FastAPI**, **DynamoDB**, and **InfluxDB** instead of stuffing raw building data into the prompt.
- Implemented an **MCP** (Model Context Protocol) agent for live point-condition time series: catalog lookup in **DynamoDB**, per-point **Influx** queries and prompt-led date windows.
- Built a **Neo4j** knowledge graph of HVAC equipment hierarchy (Equipment, Location, FEEDS, HAS_LOCATION), with **APOC**/full-text tools so the agent can search and traverse plant → AHU → VAV without writing raw **Cypher**.
- Added document **RAG** over building functional docs and onboarding FAQs to ground fault explanations in site-specific knowledge.
- Containerized **Neo4j** with the API stack and auto-loaded the equipment graph on server startup so the agent has a ready graph in **Docker**.

REVE Systems

Jan 2024 – Dec 2025

Machine Learning Engineer

- Built a **RAG** chatbot for REVE Chat using **GraphRAG**, **LightRAG**, multimodal **RAG**, agentic **RAG**, and **LangChain**, with hybrid search (**BM25** + dense) for stronger responses.
- Fine-tuned **wav2vec 2.0** and **Whisper** for Bengali **STT** (Kotha) with **PEFT-LoRA**; evaluated on multiple test sets, including augmented data, to find model weaknesses.
- Fine-tuned a **VITS** model for Bengali **TTS** (Uccharon).
- Developed a voice-based banking agent with an **STT**-chatbot-**TTS** pipeline to automate customer conversations.
- Compressed models with quantization and **ONNX** for faster load/inference and offline use.
- Implemented face liveness detection for the NTMC biometric authentication project using **OpenCV**, **CNNs**, and **MediaPipe** (anti-spoofing).
- Built time-series forecasting pipelines for production quantity and sales.
- Developed a chatbot with **BERT** and **Bi-LSTM**, and improved the REVE Chat dashboard frontend (**React.js**).

Codelab FZC

Jun 2023 – Jan 2024

Software Engineer

- Developed a 2D object-detection game and a typing game (players type sentences against a time limit) in **Unity** for Android and iOS.
- Contributed to an admin panel, implementing CRUD operations in **Laravel**.

TECHNICAL SKILLS & TECHNOLOGIES

Languages	Python, Java, C, C++, PHP, C#
Agentic AI	LangGraph, LangChain, LlamaIndex, MCP, Agentic RAG, OpenAI, Groq
Machine Learning	PyTorch, TensorFlow, OpenCV, Scikit-learn, MLflow, CI/CD
Web Technologies	Django, FastAPI, JavaScript (React.js, Node.js, Vue.js), HTML/CSS, Laravel
Databases	MySQL, MongoDB, Qdrant, Memgraph, Neo4j, InfluxDB, DynamoDB, Firebase
Cloud / Infra	AWS, Docker, Linux, Git, Bash
Tools	L ^A T _E X, Microsoft Office
Problem Solving	600+ problems — Codeforces (partha361), LeetCode (ppartha), LightOJ (parthaghoshu), StopStalk (ppaartha)

RESEARCH EXPERIENCE

- **Development of a Large Corpus of Spontaneous Speech for Improved Speech Recognition in Bangladeshi Bangla:** Created a 200-hour Bangladeshi Bangla spontaneous speech corpus and fine-tuned multilingual **wav2vec 2.0**, comparing it with a reading corpus (**SUBAK.KO**) to improve **ASR**.
- **BanSpeech: A Multi-Domain Bangla Speech Recognition Benchmark Toward Robust Performance in Challenging Conditions:** Assessed a Bangla **ASR** model on **BanSpeech**, a 7.2-hour benchmark; it struggled with spontaneous speech and high OOV rates, highlighting domain selection.
Paper: ieeexplore.ieee.org/document/10453554.

PROJECTS

- **SUST Student Management System** 
Managed student and faculty data, departments, courses, and assignments; faculty recorded student marks.
Technologies: Django, Python, HTML, CSS, Bootstrap, MySQL
- **Automatic Attendance System Using Face Recognition** 
Face-recognition web app that marks attendance from five enrolled photos per classmate.
Technologies: Python, OpenCV, Streamlit
- **Shoesly — E-commerce Website** 
Distinct user, supplier, admin, and bank flows for catalog filtering, order confirmation, and transactions.
Technologies: Node.js, React.js, MongoDB, Express.js

ACHIEVEMENTS

- 1st Runner-up, AI for Bangla 2.0 (2023).

EXTRA-CURRICULAR ACTIVITIES

CSE Society, SUST

General Secretary, Assistant General Secretary, Executive Member 2019 – 2023

SUST ACM Student Chapter

Joint Secretary 2021 – 2023

REFERENCE

Dr. Mohammad Shahidur Rahman

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